

Background

DTU - Multiple genes isoforms can result in the dominant transcript being different across samples when short-reading RNA. Differential transcript usage (DTU) is looking at these differences across samples.

Equivalence class - Many reads are compatible with several transcripts, so reads are assigned an equivalence class.

Using equivalence class counts for fast and accurate testing of differential transcript usage (Cmero et al.)

Goal: Look for differential transcript usage (DTU) between samples

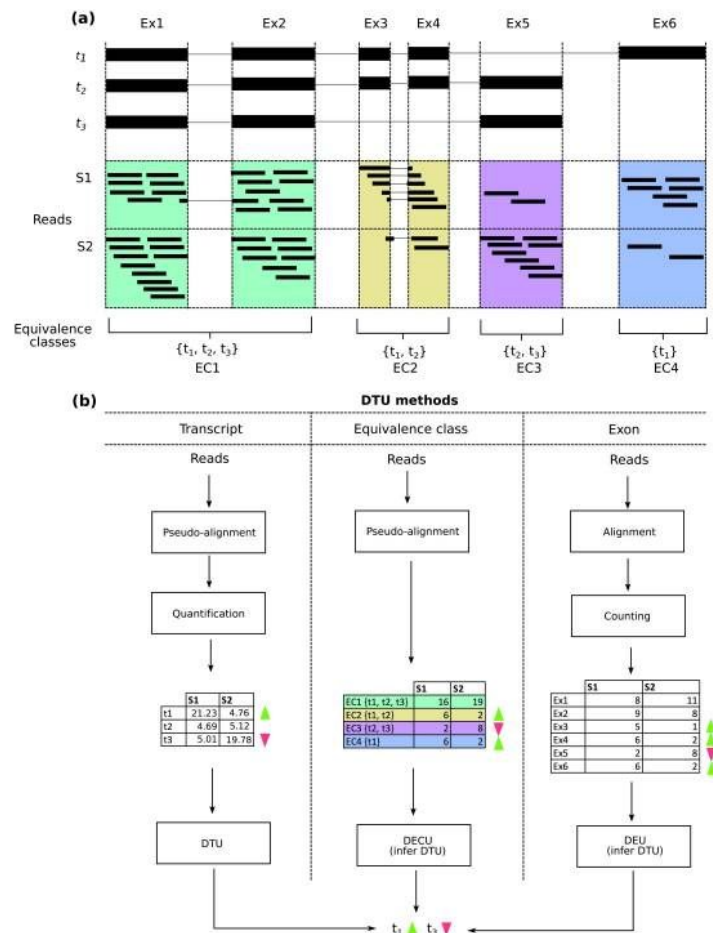
Traditional methods:

1. Exon counts
2. Directly on transcript abundances (much faster, slightly worse accuracy)

Solution: Use equivalence class counts (ECCs) to directly detect DTU

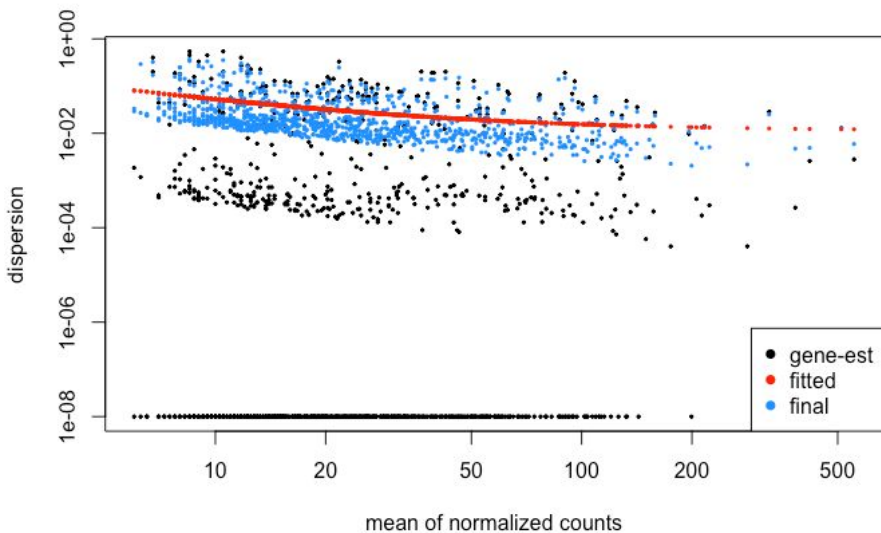
Our Process:

- 1) Replicate the figures in the paper
- 2) Run pipeline described in paper to do DTU analysis

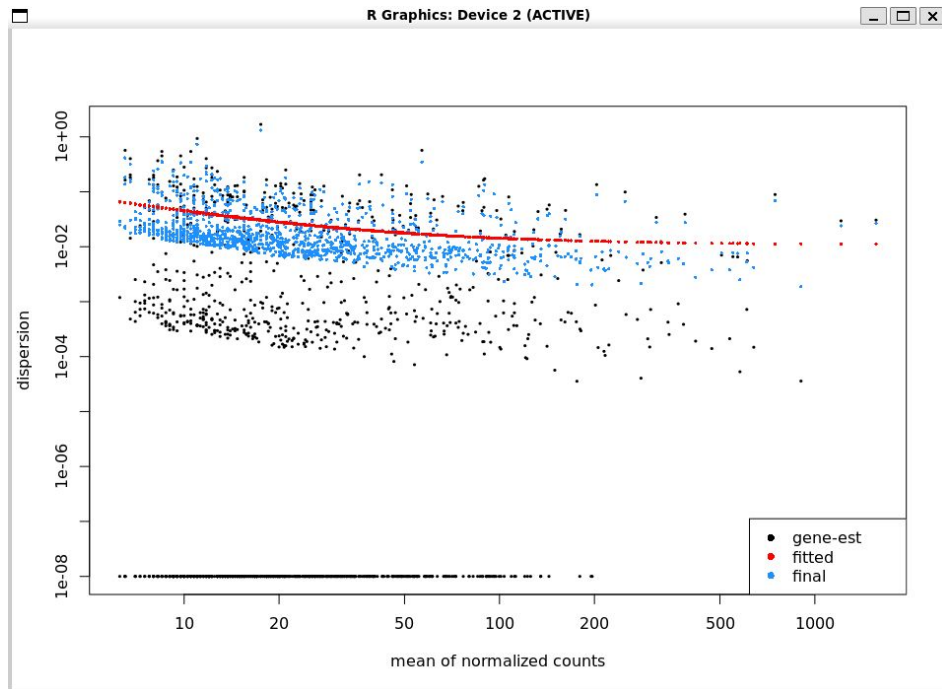


Results Comparison - Vignette Figures

Original

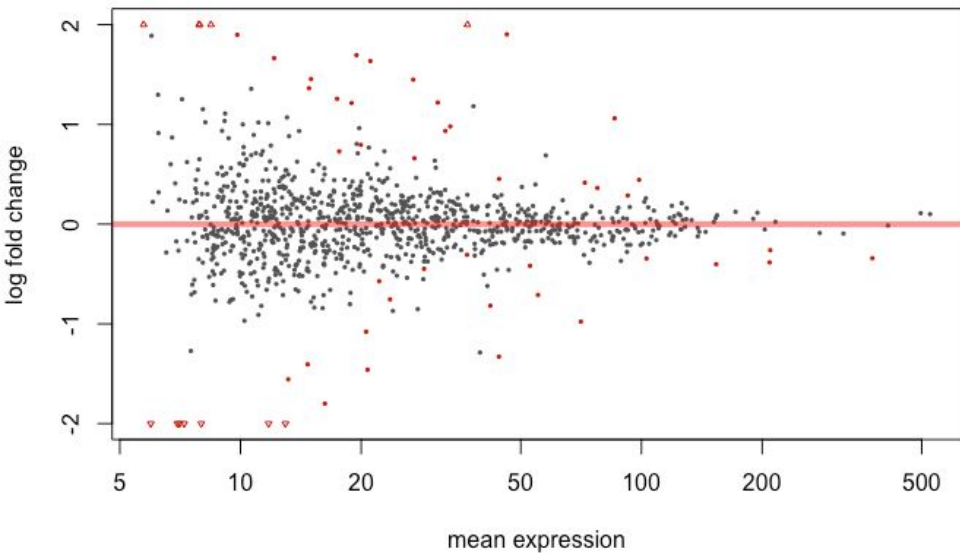


Ours

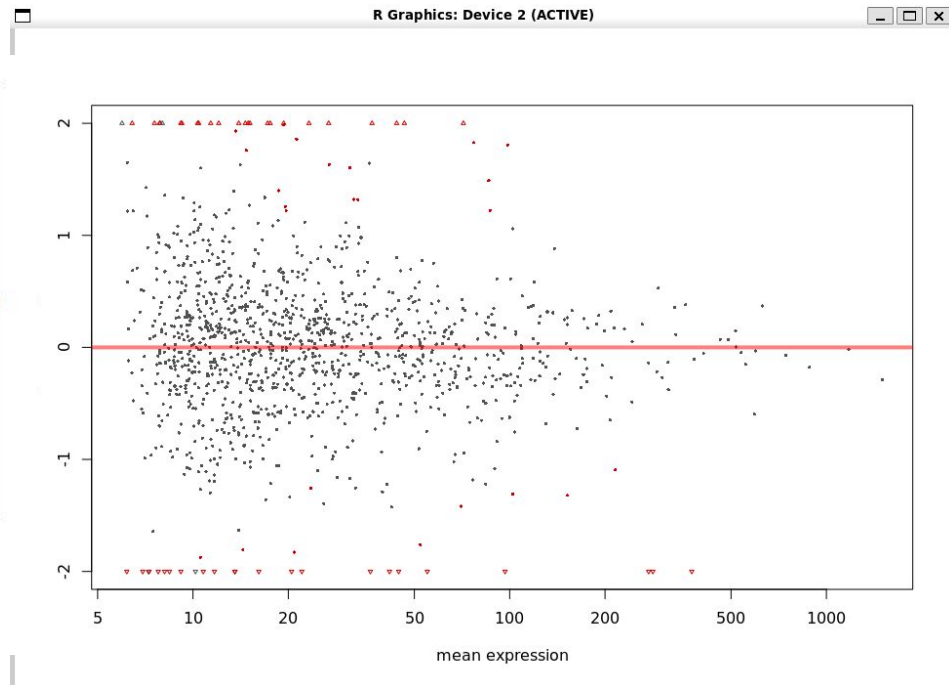


Results Comparison - Vignette Figures

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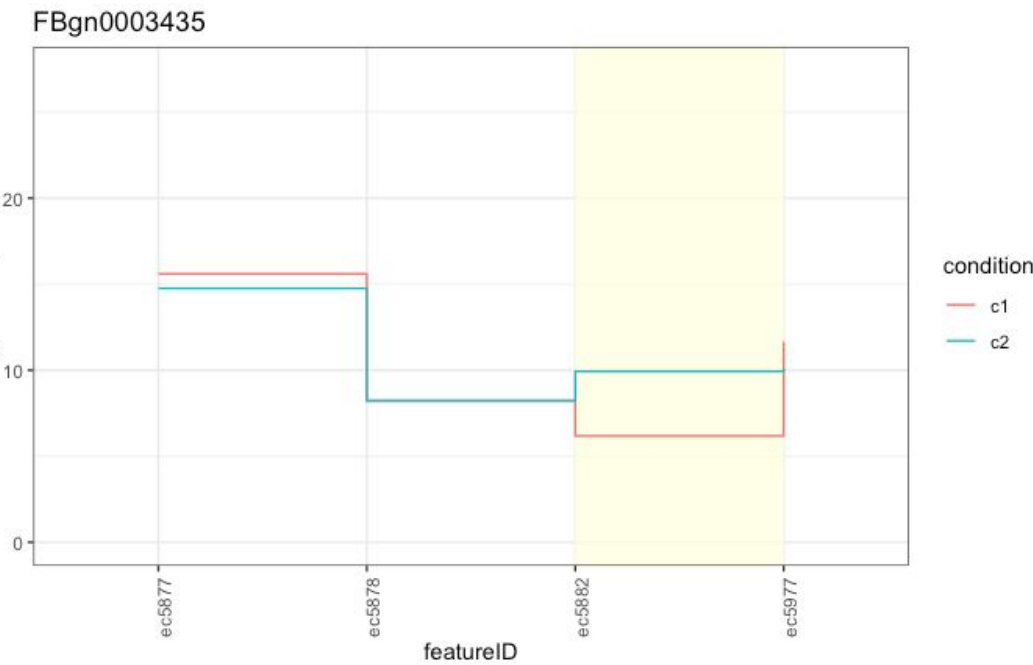


Ours

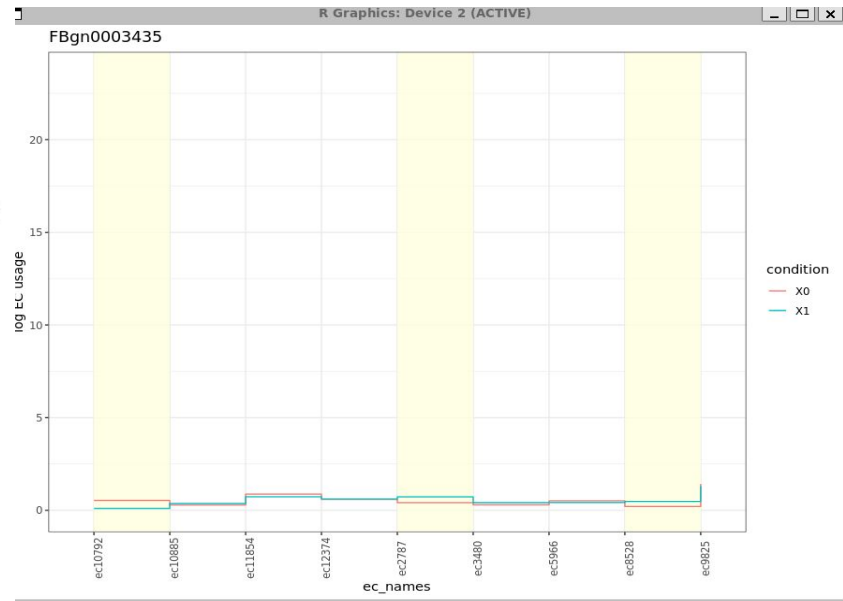


Results Comparison - Vignette Figures

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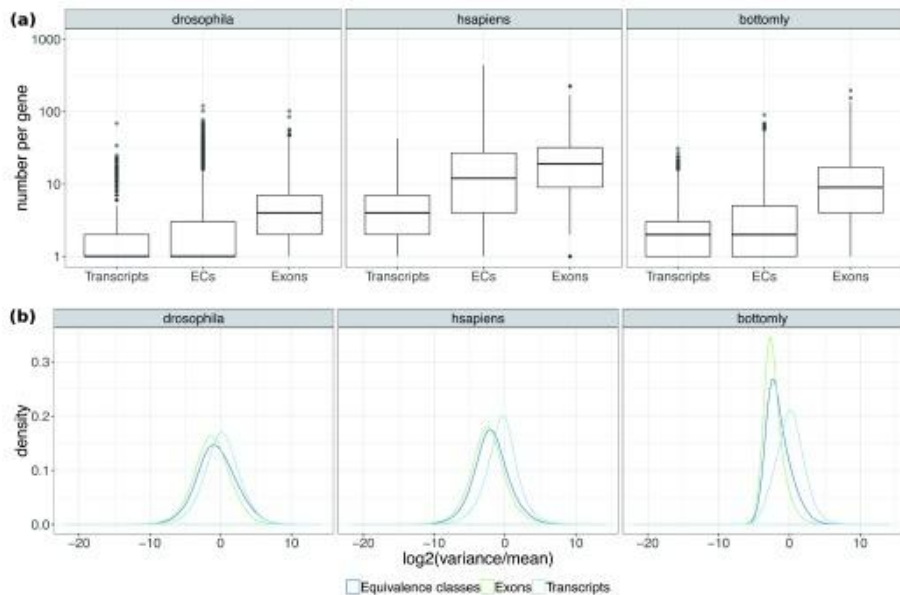


Ours

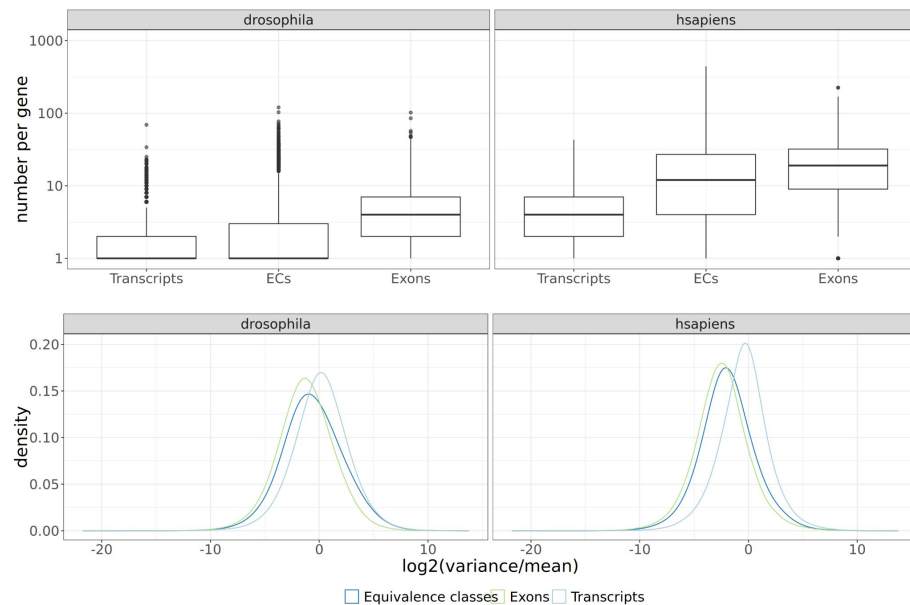


Results Comparison - Paper Figures

Original

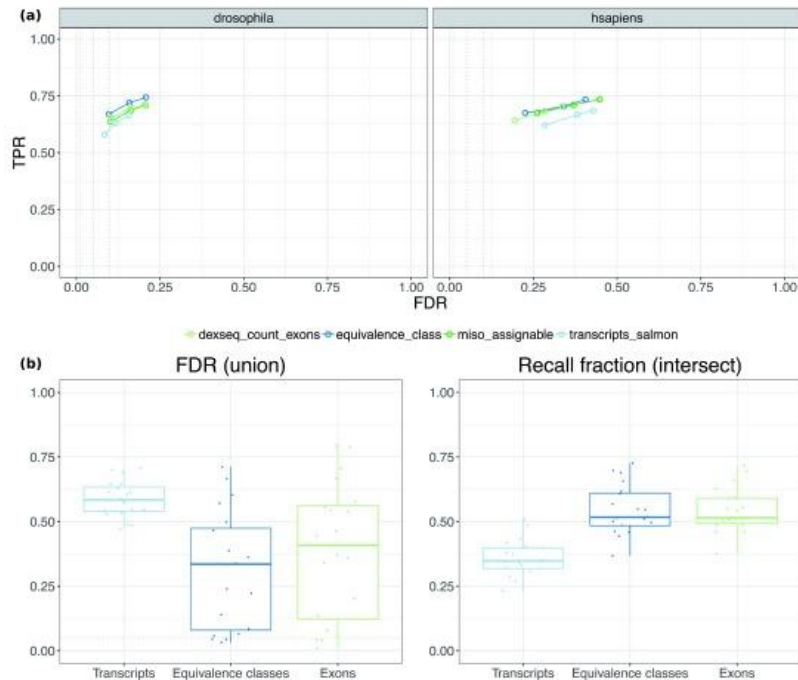


Ours



Results Comparison - Paper Figures

Original



Ours

